

Human Reproduction 人类生殖

Year 9 科学 · 考前冲刺 · 2 小时，中间不休息

0:00	开场：Do Now 和核心思想	第 1–3 页
0:04	Loop 1 · 性细胞和男女生殖系统	第 4–14 页
0:26	Loop 2 · 激素和月经周期	第 15–24 页
0:49	Loop 3 · 受精、染色体、双胞胎	第 25–35 页
1:12	Loop 4 · 怀孕、出生、不孕	第 36–49 页
1:42	失分点和总结	第 50–51 页
1:44	Mini mock (In-class work 的 Part E)	第 52 页
1:59	Homework	第 53 页

Teaching line, not a question slide.

Plan: opening 0:00, Loop 1 0:04, Loop 2 0:26, Loop 3 0:49, Loop 4 1:12, lose marks and summary 1:42, mini mock 1:44, homework 1:59.

Each loop ends with about 6 minutes of writing on the In-class work sheet (Parts A to D). Part E is the mini mock.

Say it:

reproduction = ree-pruh-DUK-shun

hormones = HOR-mohnz

menstrual = MEN-stroo-ul

fertilisation = fer-tih-ly-ZAY-shun

chromosomes = KROH-muh-sohmz

pregnancy = PREG-nun-see

fertility = fer-TIL-ih-tee

Do Now 课前小测

- 1 说出男性和女性的 sex cell (性细胞)。
male: sperm (精子); female: egg / ovum (卵)
- 2 哪个器官产生女性的性细胞？
the ovaries (卵巢)
- 3 两个性细胞结合，叫什么？
fertilisation (受精)
- 4 人的一个体细胞里有多少条 chromosomes (染色体)？
46 (23 pairs, 23 对)
- 5 哪个器官把氧气从母亲传给 foetus (胎儿)？
the placenta (胎盘)

1. sperm (male); egg or ovum (female)
 2. the ovaries
 3. fertilisation
 4. 46 (23 pairs)
 5. the placenta
- Diagnostic:

1 to 3 wrong: go slowly through Loop 1. 4 wrong: spend longer on the chromosome slide in Loop 3. 5 wrong: spend longer on the placenta slides in Loop 4.

Say it:

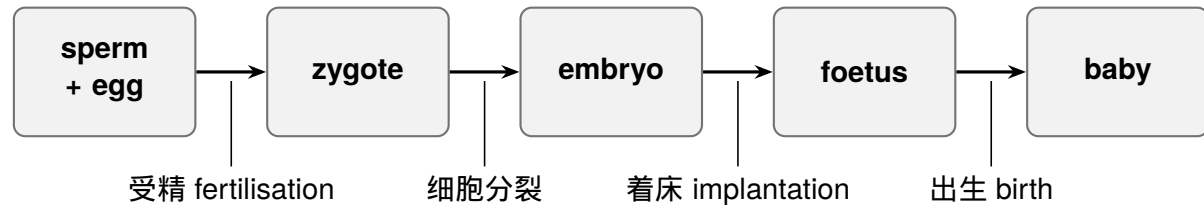
ovum = OH-vum

ovaries = OH-vuh-reez

placenta = pluh-SEN-tuh

foetus = FEE-tus

The one idea 核心思想：从两个细胞到一个婴儿



每道题都在问这条线上的某一步。

每一步要知道三样：是哪个细胞、在哪里、叫什么过程。

哪一步发生在 oviduct (输卵管) 里？

fertilisation happens in the oviduct; zygote 一边分裂一边向 uterus (子宫) 移动

Teaching line, not a question slide.

Q fertilisation happens in the oviduct; the zygote starts to divide there on its way to the uterus

Say it:

zygote = ZY-goht

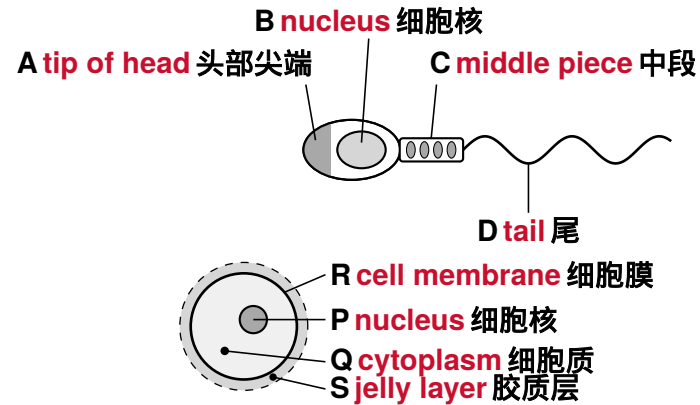
embryo = EM-bree-oh

implantation = im-plan-TAY-shun

oviduct = OH-vih-dukt

uterus = YOO-tuh-rus

Quick review 快速回忆：性细胞



- 1 说出 A–D 和 P–S 的名称。
见图上红字
- 2 每个 nucleus 里有多少条 chromosomes ?
23 in each
- 3 哪个细胞更大？哪个会动？
**the egg is much larger;
only the sperm can move:
it swims with its tail**

1. A tip of the head (it holds enzymes), B nucleus, C middle piece (it holds mitochondria), D tail; P nucleus, Q cytoplasm, R cell membrane, S jelly layer
2. 23 in each
3. the egg is much larger; only the sperm can move: it swims with its tail

Say it:

nucleus = NYOO-klee-us

cytoplasm = SY-toh-plaz-um

membrane = MEM-brayn

enzymes = EN-zymz

mitochondria = my-toh-KON-dree-uh

Adaptations 适应：特征 + 有什么用

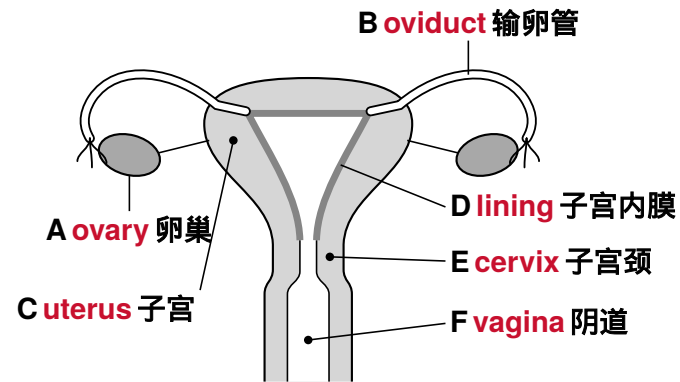
细胞	Feature 特征	How it helps 有什么用
sperm	long tail 长尾	<u>it swims to the egg</u>
sperm	很多 mitochondria	<u>they release energy for swimming</u>
sperm	头部有 enzymes	<u>they break through the jelly layer and membrane</u>
egg	cytoplasm 很大	<u>a food store for the first cell divisions</u>
egg	membrane 会变化	<u>it stops any other sperm getting in</u>

为什么男性产生几百万个 sperm，而女性每月只排一个 egg ?
very few sperm survive the journey to the oviduct, so millions give a better chance that one meets the egg

Teaching line, not a question slide.
Gaps: it swims to the egg; they release energy for swimming; they break through the jelly layer and membrane of the egg; it is a food store for the first cell divisions; it stops any other sperm getting in.

Q very few sperm survive the journey to the oviduct, so millions give a better chance that one meets the egg; one egg is enough because it stays inside the body
Say it:
adaptations = ad-ap-TAY-shunz

Quick review 快速回忆：女性生殖系统，正面图



- 1 说出 A–F 的名称。
见图上红字。B = fallopian tube; D = endometrium
- 2 卵在哪里产生？
in the ovaries
- 3 胎儿在哪里发育？
in the uterus
- 4 哪一部分是一圈肌肉？
the cervix

1. A ovary, B oviduct (fallopian tube), C uterus, D lining of the uterus (endometrium), E cervix, F vagina
2. in the ovaries
3. in the uterus
4. the cervix

Say it:

ovary = OH-vuh-ree

fallopian = fuh-LOH-pee-un

endometrium = en-doh-MEE-tree-um

cervix = SUR-viks

vagina = vuh-JY-nuh

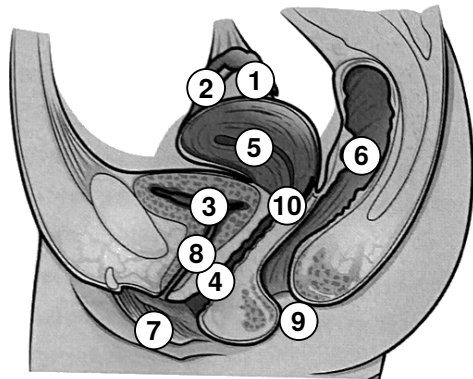
女性生殖系统：每部分的作用

Part 部位	Job 作用
ovary 卵巢	产生 <u>eggs</u> 和两种 hormones (oestrogen、 progesterone)
oviduct 输卵管	运送 egg； <u>fertilisation</u> 在这里发生
uterus 子宫	<u>foetus</u> 在这里发育
lining 子宫内膜	变厚，让 <u>embryo</u> 可以 implant (着床)
cervix 子宫颈	uterus 底部的一圈 <u>muscle</u>
vagina 阴道	接收 <u>sperm</u> (semen)； 婴儿从这里出生

Teaching line, not a question slide.
Gaps: eggs; fertilisation; foetus; embryo; muscle; sperm (semen).
The two hormones from the ovary are oestrogen and progesterone.

Say it:
oestrogen = EE-struh-jun
progesterone = proh-JES-tuh-rohn
semen = SEE-mun

Quick review 快速回忆：女性生殖系统，侧面图



anus · bladder · cervix · fallopian tube · ovary ·
rectum · urethra · uterus · vagina · vulva

1 把这些词和 1-10 对上。

1 ovary 卵巢

2 fallopian tube 输卵管

3 bladder 膀胱

4 vagina 阴道

5 uterus 子宫

6 rectum 直肠

7 vulva 外阴

8 urethra 尿道

9 anus 肛门

10 cervix 子宫颈

2 哪四个不是生殖器官？

**bladder, urethra (urine 尿); rectum, anus
(消化系统)**

1. 1 ovary, 2 fallopian tube, 3 bladder, 4 vagina, 5 uterus, 6 rectum,
7 vulva, 8 urethra, 9 anus, 10 cervix

2. bladder and urethra (they store and carry urine); rectum and
anus (digestive system)

Say it:

anus = AY-nus

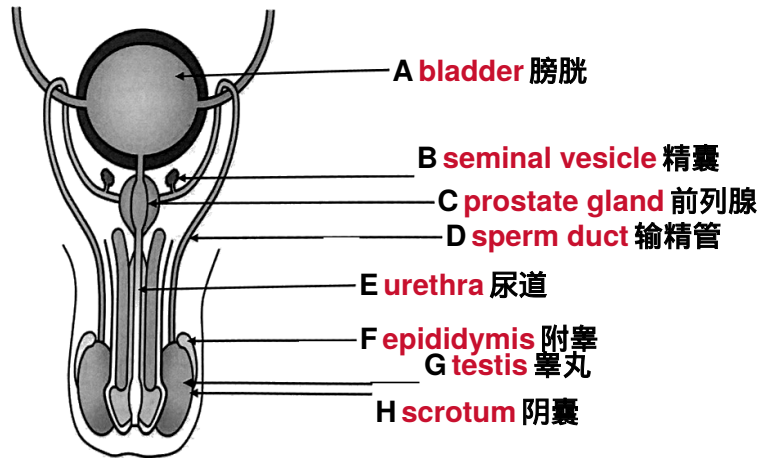
rectum = REK-tum

urethra = yoo-REE-thruh

vulva = VUL-vuh

urine = YOOR-in

Quick review 快速回忆：男性生殖系统



- 1 说出 A-H 的名称。
见图上红字。D = **vas deferens**
- 2 sperm 在哪里产生？
in the testes
- 3 哪条管道既走 urine 又走 semen？
the urethra

1. A bladder, B seminal vesicle, C prostate gland, D sperm duct (vas deferens), E urethra, F epididymis, G testis, H scrotum

2. in the testes

3. the urethra

Say it:

seminal = SEM-ih-nul

vesicle = VES-ih-kul

prostate = PROS-tayt

vas deferens = vas-DEF-uh-renz

epididymis = ep-ih-DID-ih-mis

testis = TES-tis

testes = TES-teez

scrotum = SKROH-tum

一个 sperm cell 走的路

把卡片排好顺序。

A urethra 把 semen 经 penis 送出体外。

B sperm 在 testis 里产生。

C sperm 沿着 sperm duct 走。

D sperm 储存在 epididymis 里。

E glands (腺体) 加入液体: sperm + fluid = **semen**。

顺序: B → D → C → E → A

为什么 testes 要放在体外？

sperm are made best a little below body temperature; in the scrotum the testes stay cooler

Teaching line, not a question slide.

Order: B, D, C, E, A. Gap: semen.

Q sperm are made best at a temperature a little below body

temperature; in the scrotum the testes stay cooler

Say it:

penis = PEE-nis

男性生殖系统：每部分的作用

Part 部位	Job 作用
scrotum 阴囊	把 testes 放在体外，让它们保持 <u>cooler</u>
testes 睾丸	产生 <u>sperm</u> 和激素 <u>testosterone</u>
epididymis 附睾	<u>stores</u> sperm
sperm duct 输精管	把 sperm 运到 <u>urethra</u>
glands 腺体	加入 <u>fluid</u> ，给 sperm 提供养分
penis 阴茎	把 semen 送进 <u>vagina</u>

Teaching line, not a question slide.
Gaps: cooler; sperm, testosterone; stores; urethra; fluid; vagina.

The glands are the seminal vesicles and the prostate gland.
Say it:
testosterone = tes-TOS-tuh-rohn

Spot the error 找错 (1)

题目：说出性细胞在哪里产生，胎儿在哪里发育。

Sam's answer

Eggs are made in the uterus and travel down the oviduct. The cervix is the place where the foetus develops. In a man, sperm are made in the testes and stored in the scrotum.

Sam 的答案有 3 处错。找出来，改正，并说出为什么丢分。

1. eggs are made in the ovaries, not in the uterus (L4)
2. the foetus develops in the uterus; the cervix is the ring of muscle at its base (L4)
3. sperm are stored in the epididymis; the scrotum is the bag of skin that holds the testes (L4)

1. eggs are made in the ovaries, not in the uterus (L4)
2. the foetus develops in the uterus; the cervix is the ring of muscle at its base (L4)
3. sperm are stored in the epididymis; the scrotum is the bag of skin

that holds the testes (L4)

Short solution:

Eggs are made in the ovaries and travel down the oviduct. The uterus is the place where the foetus develops. In a man, sperm are made in the testes and stored in the epididymis.

Practice 练习

Q1 说出 sperm cell 和 egg cell 的两个不同点。 [A]

any two: a sperm is much smaller; a sperm has a tail and swims, an egg cannot move; an egg has a large food store; millions of sperm, one egg a month

Q2 解释为什么 testes 在体外的 scrotum 里。 [M]

sperm are made best a little below body temperature; outside the body the testes stay cooler

Q3 一位女性两侧的 oviducts 都堵了，但每月仍然排卵。解释她为什么不能自然怀孕。 [E]

fertilisation happens in the oviduct; the sperm cannot reach the egg and the egg cannot reach the uterus, so no zygote forms and nothing can implant

Q1 any two: a sperm is much smaller; a sperm has a tail and can swim, an egg cannot move; an egg has a large food store in its cytoplasm; millions of sperm are made, one egg is released each month

Q2 sperm are made best at a temperature a little below body temperature; outside the body the testes stay cooler

Q3 fertilisation happens in the oviduct; the sperm cannot reach the egg and the egg cannot reach the uterus, so no zygote forms and nothing can implant

In-class work · Part A (约 6 分钟)

- 1 给女性生殖系统的图标注。
- 2 补全关于男性生殖系统的句子。
- 3 解释 sperm cell 的一个 adaptation。

要求：写完整的句子，用科学词。

答案在 inclass_answers.pdf 的 Part A

Puberty 青春期：由 hormones 启动

Hormones (激素) 是化学 messengers，随 blood 运到全身。

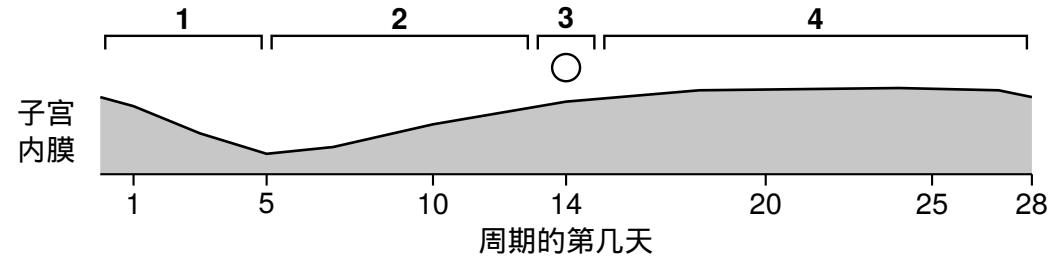
	男孩	女孩
hormone 激素	<u>testosterone</u>	<u>oestrogen</u>
产生于	testes	ovaries
变化	声音变低	<u>periods</u> 开始
另外	开始产生 sperm	乳房发育

判断对错：女孩通常比男孩先进入 puberty。 **True**

Teaching line, not a question slide.
Gaps: messengers; blood; testosterone; oestrogen; periods.
Q true

Both sexes: a growth spurt, and pubic and underarm hair.
Say it:
puberty = PYOO-buh-tee

Menstrual cycle 月经周期：第 1、2 阶段



1 第 1–5 天：内膜 **breaks down**，经 vagina 流出。这就是 **period (menstruation)**。

2 第 5–13 天：内膜重新 **builds up**；一个 egg 在 **ovary** 里成熟。

阶段名称：1 menstrual 2 follicular

Teaching line, not a question slide.

Gaps: breaks down; period (menstruation); builds up; ovary.

Stage names: 1 menstrual, 2 follicular. In stage 2 a follicle in the ovary grows; FSH (follicle stimulating hormone) makes the egg inside it mature.

Say it:

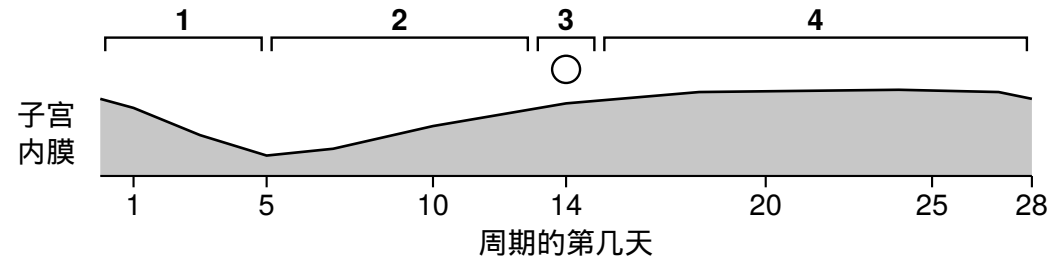
menstruation = men-stroo-AY-shun

follicular = fuh-LIK-yoo-luh

follicle = FOL-ih-kul

stimulating = STIM-yoo-lay-ting

Menstrual cycle 月经周期：第 3、4 阶段



3 大约第 14 天：一个 egg 从 ovary **released**。这叫 **ovulation** (排卵)。

4 第 15–28 天：内膜保持 **thick**，等待受精卵。

阶段名称：3 ovulation 4 luteal。没有受精卵：内膜 breaks down，周期重新开始

她什么时候最容易怀孕？

around day 14: the day of ovulation and the two days before and after it

Teaching line, not a question slide.

Gaps: released; ovulation; thick.

Q around day 14: on the day of ovulation and the two days before and after it

Stage names: 3 ovulation, 4 luteal. With no fertilised egg the lining breaks down and the cycle starts again.

Say it:

ovulation = ov-yoo-LAY-shun

luteal = LOO-tee-ul

四种 hormones，四个作用

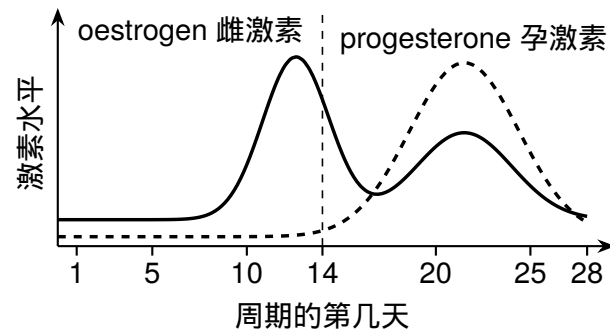
Teaching line, not a question slide.
Gaps: mature; pituitary gland, ovulation; builds up; ovary, thick.
Q FSH and LH

Say it:
pituitary = pih-TYOO-ih-tree
luteinising = LOO-tee-in-eye-zing

Hormone	产生于	作用
FSH	pituitary gland (垂体)	让一个 egg 在 ovary 里 <u>mature</u>
LH	<u>pituitary gland</u>	在第 14 天触发 <u>ovulation</u>
oestrogen	ovary	<u>builds up</u> 子宫内膜
progesterone	<u>ovary</u>	让内膜保持 <u>thick</u>

FSH = follicle stimulating hormone。LH = luteinising hormone。
哪两种激素在 ovulation 时达到峰值？ **FSH and LH**

读图：ovaries 产生的激素



- 1 哪种激素在第 14 天之前达到峰值？
oestrogen
- 2 第 21 天左右哪种激素最高？
progesterone
- 3 到第 28 天两种都下降了。内膜会怎样？
the lining breaks down and is lost: a period starts

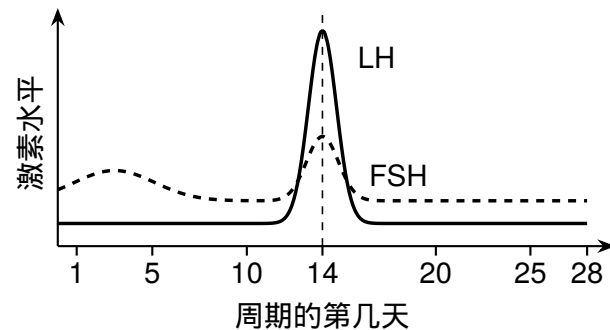
Teaching line, not a question slide.

1. oestrogen

2. progesterone

3. the lining breaks down and is lost: a period starts

读图：pituitary gland 产生的激素



- 1 LH 和 FSH 在第几天达到峰值？
day 14
- 2 LH 的峰值引起什么？
ovulation: an egg is released from an ovary
- 3 描述第 1–12 天 LH 的水平。
it stays low and almost constant

Teaching line, not a question slide.

1. day 14

2. ovulation: an egg is released from an ovary

3. it stays low and almost constant

卵没有受精：按这个顺序写

Teaching line, not a question slide.

Gaps: implants; progesterone; breaks down; vagina.

Q the embryo implants, progesterone stays high and the lining stays thick, so there is no period

- 1 卵没有受精，所以没有 embryo implants 到内膜里。
- 2 progesterone 的水平下降。
- 3 子宫内膜 breaks down。
- 4 血和内膜经 vagina 排出：这就是 period。

A: 第 3–4 行 **M:** 把激素和内膜连起来 **E:** 整条链

如果卵受精了，有什么不同？

the embryo implants, progesterone stays high and the lining stays thick, so there is no period

Spot the error 找错 (2)

题目：描述 ovulation，并说出一种改变子宫内膜的激素。

Sam's answer

Ovulation is when the lining of the uterus is lost. An egg is released on about day 28 of the cycle. Oestrogen is made in the pituitary gland and makes the lining thicker.

Sam 的答案有 3 处错。找出来，改正，并说出为什么丢分。

1. ovulation is the release of an egg from an ovary; losing the lining is menstruation (L5)
2. the egg is released on about day 14, not day 28 (L5)
3. oestrogen is made in the ovaries, not in the pituitary gland (L6)

Short solution:

Ovulation is the release of an egg from an ovary, on about day 14 of the cycle. Oestrogen is made in the ovaries and makes the lining thicker.

1. ovulation is the release of an egg from an ovary; losing the lining is menstruation (L5)
2. the egg is released on about day 14, not day 28 (L5)
3. oestrogen is made in the ovaries, not in the pituitary gland (L6)

Practice 练习

- Q1** 一位女性的周期是 28 天。她在哪几天最容易怀孕？ **[A]**
about days 12 to 16: ovulation is on about day 14, and she is fertile for two days before and after it
- Q2** 描述第 5 天到第 14 天子宫内膜的变化，并说出引起变化的激素。 **[M]**
the lining builds up and becomes thicker, with more blood vessels; oestrogen
- Q3** 孕妇的 progesterone 水平一直很高。解释为什么这很重要。 **[E]**
progesterone keeps the lining thick, so it is not lost; the embryo stays implanted and the placenta can form, so there are no periods during pregnancy

Q1 about days 12 to 16: ovulation is on about day 14, and she is fertile for two days before and after it
Q2 the lining builds up and becomes thicker, with more blood

vessels; oestrogen
Q3 progesterone keeps the lining of the uterus thick, so the lining is not lost; the embryo stays implanted and the placenta can form, so there are no periods during pregnancy

In-class work · Part B (约 6 分钟)

- 1 说出周期的四个阶段。
- 2 把每种激素和它的作用连起来。
- 3 读一张激素水平的数据表并解释。

数据题：先描述趋势，带上数字。

答案在 inclass_answers.pdf 的 Part B

从 intercourse 到 fertilisation

把卡片排好顺序。

A sperm 游过 cervix 和 uterus，进入 oviducts。

B penis 充血，变硬（erect）。

C 一个 sperm 进入 egg，两个 nuclei（细胞核）融合。

D semen 被射入 vagina。这叫 ejaculation（射精）。

E penis 放入 vagina。

顺序：B → E → D → A → C

哪张卡片是 fertilisation？它在哪里发生？

card C; it happens in the oviduct

Teaching line, not a question slide.

Order: B, E, D, A, C.

Q card C is fertilisation; it happens in the oviduct

Say it:

intercourse = IN-ter-kors

ejaculation = ih-jak-yoo-LAY-shun

Zygote、embryo、foetus

Teaching line, not a question slide.

Gaps: zygote; embryo; implantation; foetus.

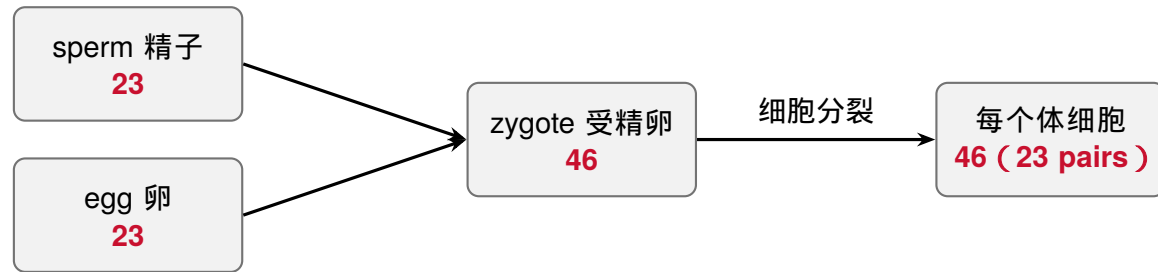
Q the membrane of the egg changes as soon as one sperm has entered

- 1 sperm 的 nucleus 和 egg 的 nucleus 融合。这个新细胞叫 zygote (受精卵)。
- 2 它沿着 oviduct 移动，同时不断分裂。这团细胞叫 embryo (胚胎)。
- 3 这团细胞陷进子宫内膜。这叫 implantation (着床)。
- 4 大约 8 周后，器官都有了。这时叫 foetus (胎儿)。

什么阻止第二个 sperm 进入 egg ?

the membrane of the egg changes as soon as one sperm has entered

Chromosomes 染色体：数一数



Chromosomes 带着 genes (基因), 在每个细胞的 nucleus 里。

为什么 sex cell 只能有一半的数目？

so that the zygote has the full 46 and not 92 when the two sex cells join; half comes from each parent

Teaching line, not a question slide.

Gaps: sperm 23; egg 23; zygote 46; every body cell 46 (23 pairs); nucleus.

Q so that the zygote has the full 46 and not 92 when the two sex cells join; half comes from each parent

Say it:
genes = JEENZ

生男还是生女？

		母亲的 eggs	
		X	X
父亲的 sperm	X	XX	XX
	Y	XY	XY

女性的细胞是 XX，男性的细胞是 XY。
每个 egg 都带 X。sperm 带 X 或 Y。

- 1 把方格填完。
见左图红字
- 2 生男孩的概率是多少？
50 % (2 of the 4 boxes)
- 3 哪一方的性细胞决定性别？
the father's: a sperm with X gives a girl, a sperm with Y gives a boy

Teaching line, not a question slide.

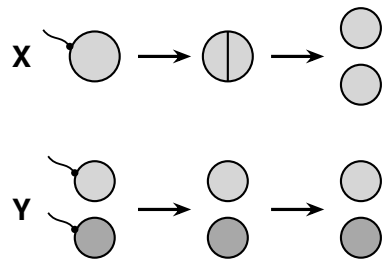
Gaps: X; X or Y.

1. top row XX, XX; bottom row XY, XY

2. 50 % (2 of the 4 boxes)

3. the father's: a sperm with X gives a girl, a sperm with Y gives a boy

Twins 双胞胎



每一行最后都是两个 embryos。

1 哪一行是 identical twins (同卵双胞胎)?

row X

2 X 行: **one** 个 egg, **one** 个 sperm. zygote **splits into two**。

3 Y 行: **two** 个 eggs, **two** 个 sperm。

4 哪种双胞胎可以是一男一女?

row Y, the non-identical (fraternal) twins: they have different genes. Identical twins have the same genes, so they are always the same sex.

Teaching line, not a question slide.

1. row X

2. one, one; splits into two

3. two, two

4. row Y, the non-identical (fraternal) twins: they have different genes; identical twins have the same genes, so they are always the same sex

Say it:

identical = eye-DEN-tih-kul

fraternal = fruh-TUR-nul

Workbook 练习册：对还是错？

- a** 同卵双胞胎：一个受精卵分裂成两个，而且两个细胞分开了。 **True**
- b** 同卵双胞胎一定是不同性别。 **False: always the same sex**
- c** 异卵双胞胎：两个卵细胞都受精了。 **True**
- d** 异卵双胞胎可以是同性别，也可以是不同性别。 **True**
- e** 同卵双胞胎：两个卵细胞受精。 **False: that gives non-identical twins**
- f** 异卵双胞胎：卵细胞自己变成胚胎。 **False: each egg must be fertilised by a sperm**

- (a) true
- (b) false: they are always the same sex
- (c) true

- (d) true
- (e) false: that gives non-identical twins
- (f) false: each egg must be fertilised by a sperm

Workbook 练习册: fertilisation

Q1 fertilisation 在哪里发生？

in the oviduct (fallopian tube)

Q2 只有一个 sperm cell 和 egg cell 结合。你认为其他的 sperm 怎样了？

they die and are broken down

Q3 人类在进化中上了陆地，变成 internal fertilisation (体内受精)。为什么这在陆地上是优势？

on land the sex cells would dry out; inside the body the sperm can swim in fluid, the sex cells are protected, and a sperm is much more likely to meet the egg

Q1 in the oviduct (fallopian tube)

Q2 they die and are broken down

Q3 on land the sex cells would dry out; inside the body the sperm

can swim in fluid, the sex cells are protected, and a sperm is much more likely to meet the egg

Say it:

evolutionary = ee-vuh-LOO-shun-ree

下一个是男孩吗？按这个顺序写

Teaching line, not a question slide.

Gaps: X; Y; 50 % (1 in 2); chance.

Each fertilisation is a new event with a new sperm cell.

一对夫妇有三个女儿。第四个孩子是男孩的概率是多少？

- 1 每个 egg 都带一条 X 染色体。
- 2 一半的 sperm 带 X，一半带 Y。
- 3 所以每一次受精，生男孩的概率都是 50 %。
- 4 以前生的孩子不会改变这个 chance。

A: 说出数字 **M:** 第 1–2 行 **E:** 再加上第 4 行

Spot the error 找错 (3)

题目：解释婴儿的染色体从哪里来，以及双胞胎怎样会是 identical 的。

Sam's answer

A sperm has 46 chromosomes and an egg has 46, so the baby gets all of them. The mother's egg decides if the baby is a boy or a girl. Identical twins form when two eggs are fertilised by two sperm.

Sam 的答案有 3 处错。找出来，改正，并说出为什么丢分。

1. a sperm and an egg have 23 chromosomes each, so the zygote has 46 (L1)
2. the sperm decides the sex: it carries X or Y, and every egg carries X (L7)
3. identical twins come from one fertilised egg that splits; two eggs give non-identical twins (L11)

1. a sperm and an egg have 23 chromosomes each, so the zygote has 46 (L1)
2. the sperm decides the sex: it carries X or Y, and every egg carries X (L7)
3. identical twins come from one fertilised egg that splits; two eggs

give non-identical twins (L11)

Short solution:

A sperm has 23 chromosomes and an egg has 23, so the zygote has 46. The father's sperm decides if the baby is a boy or a girl. Identical twins form when one fertilised egg splits into two.

Practice 练习

Q1 说出 fertilisation 在哪里发生，以及形成的细胞叫什么。 [A]

in the oviduct; a zygote

Q2 一对双胞胎是一男一女。他们是 identical twins 吗？解释。 [M]

no: identical twins come from one zygote, so they have the same chromosomes and are the same sex; a boy and a girl must come from two eggs and two sperm

Q3 解释为什么孩子有父母双方的特征，但和任何一方都不完全一样。 [E]

23 chromosomes come from the mother's egg and 23 from the father's sperm, so the child's genes are a mixture of both; the mixture is a new combination, different from each parent

Q1 in the oviduct; a zygote

Q2 no: identical twins come from one zygote, so they have the same chromosomes and are the same sex; a boy and a girl must

come from two eggs and two sperm

Q3 the child gets 23 chromosomes from the mother's egg and 23 from the father's sperm, so its genes are a mixture of both; the mixture is a new combination, different from each parent

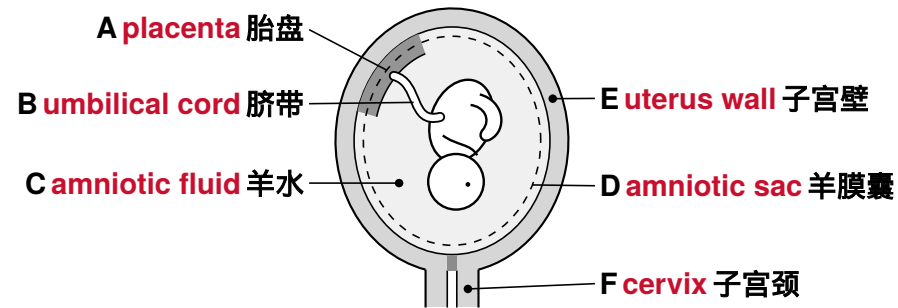
In-class work · Part C (约 6 分钟)

- 1 把从性细胞到 foetus 的各阶段排序。
- 2 根据染色体判断婴儿的性别。
- 3 解释两种双胞胎是怎样形成的。

每次提到一种细胞，都写出它的染色体数目。

答案在 inclass_answers.pdf 的 Part C

Quick review 快速回忆：子宫里的 foetus

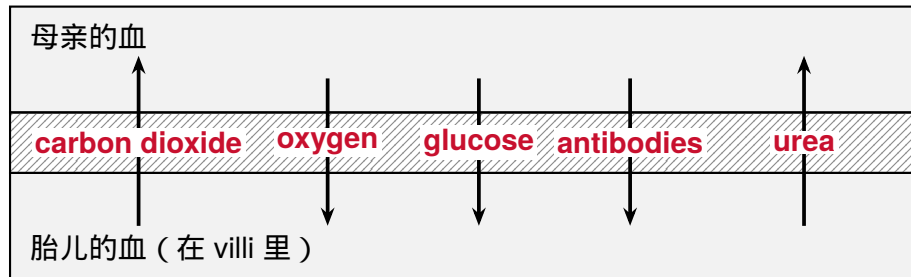


- 1 说出 A-F 的名称。
见图上红字
- 2 哪一部分给胎儿缓冲？
the amniotic fluid
- 3 哪一部分把胎儿和 placenta 连起来？
the umbilical cord

1. A placenta, B umbilical cord, C amniotic fluid, D amniotic sac, E uterus wall, F cervix
2. the amniotic fluid

3. the umbilical cord
Say it:
umbilical = um-BIL-ih-kul
amniotic = am-nee-OT-ik

Placenta 胎盘：什么能通过？



向下的箭头：胎儿需要的三样东西。向上的箭头：两种废物。

两边的血永远不 **mix**。物质靠 **diffusion**（扩散）通过。

glucose 也可以答 food / nutrients; antibodies 是抗体; urea 是尿素

Teaching line, not a question slide.

Down: oxygen; glucose (food, nutrients); antibodies. Up: carbon dioxide; urea.

Gaps: mix; diffusion.

Say it:

villi = VIL-eye

glucose = GLOO-kohz

antibodies = AN-tee-bod-eez

urea = yoo-REE-uh

diffusion = dih-FYOO-zhun

Workbook 练习册: placenta 的作用

Placenta 由胎儿的组织发育而来，让母亲和胎儿之间进行物质的 exchange (diffusion)。它有大量互不接触的血管，保证母亲的血和胎儿的血不会 mixing。

a 说出母血和胎血不能混合的三个原因。

they can be different blood types; the mother's blood pressure is higher and would damage the foetus's small blood vessels; it stops many bacteria and harmful substances reaching the foetus

b 解释为什么 embryo 需要 oxygen 和 glucose。

for respiration, which releases the energy its cells need to divide and grow

c 哪些物质从胎儿的血扩散到母亲的血？为什么必须把它们除去？

carbon dioxide and urea; they are waste products and would poison the foetus if they built up

(a) the mother and the foetus can have different blood types; the mother's blood pressure is higher and would damage the foetus's small blood vessels; keeping the bloods apart stops many bacteria and harmful substances reaching the foetus

(b) for respiration, which releases the energy its cells need to divide and grow

(c) carbon dioxide and urea; they are waste products and would

poison the foetus if they built up

Short solution:

Blanks: exchange (diffusion); mixing.

Say it:

foetal = FEE-tul

maternal = muh-TUR-nul

respiration = res-pih-RAY-shun

Oxygen 怎样到胎儿：按这个顺序写

- 1 母亲的血里 oxygen 的 concentration (浓度) 比胎儿的血 higher。
- 2 oxygen 从高浓度向低浓度 diffuses，穿过薄壁。
- 3 villi (绒毛) 提供很大的 surface area，所以扩散得快。
- 4 胎儿的血经 umbilical vein 把 oxygen 带走。

A: 第 2 行 **M:** 第 1–2 行 **E:** 再加上第 3–4 行

现在用同样的方式讲 carbon dioxide。

the foetus's blood, arriving in the umbilical artery, has a higher concentration of carbon dioxide than the mother's blood, so carbon dioxide diffuses across the villi into the mother's blood, which carries it away

Teaching line, not a question slide.

Gaps: higher; diffuses; surface area; vein.

Q the foetus's blood, arriving in the umbilical artery, has a higher concentration of carbon dioxide than the mother's blood, so carbon

dioxide diffuses across the villi into the mother's blood, which carries it away

Say it:

concentration = kon-sun-TRAY-shun

artery = AR-tuh-ree

危害：还有什么能通过 placenta ？

来自母亲	对胎儿的影响
cigarette smoke 香烟烟雾	到达胎儿的 <u>oxygen</u> 变少；婴儿偏小
alcohol 酒精	损伤发育中的 <u>brain</u>
drugs 毒品	婴儿出生时可能已经 addicted (成瘾)
rubella virus 风疹病毒	耳聋；心脏和眼睛的缺陷

为什么很少的量也会造成很大的伤害？
the foetus is very small and is growing very quickly

Teaching line, not a question slide.
Gaps: oxygen; brain.
Q the foetus is very small and is growing very quickly
Cigarette smoke: carbon monoxide takes the place of oxygen in the mother's blood; nicotine also crosses the placenta.

Say it:
rubella = roo-BEL-uh
addicted = uh-DIK-tid
monoxide = mon-OK-syd
nicotine = NIK-uh-teen

怀孕里的几个数字

- 1 怀孕 (gestation) 大约 40 周，从最后一次月经的第一天算起。
- 2 大约从第 8 周起，embryo 改叫 foetus。
- 3 20 周之前失去胎儿，叫 miscarriage (流产)。
- 4 38 周之前出生，叫 premature (早产)。

一个婴儿在 35 周出生。用哪个词描述？

premature

Teaching line, not a question slide.

Gaps: 40; 8 (the glossary says after 8 weeks, page 28 says nine);
20; 38.

Q premature

After 20 weeks the loss of a baby is called a stillbirth.

Say it:

gestation = jes-TAY-shun

miscarriage = MIS-ka-rij

premature = PREM-uh-chur

Birth 出生：三个阶段

把卡片排好顺序，再说每个阶段的名称。

A placenta 被排出。

B uterus 的肌肉收缩；cervix 张开 (dilates); 羊水破了。

C 婴儿经 vagina 被推出；脐带结扎后剪断。

顺序：B → C → A

Stage 1 labour Stage 2 delivery Stage 3 afterbirth

“the waters” 是什么？

the amniotic fluid

Teaching line, not a question slide.

Order: B, C, A. Stage 1 labour; stage 2 delivery; stage 3 afterbirth.

Q the amniotic fluid

Say it:

dilates = dy-LAYTS

labour = LAY-bur

Workbook 练习册：不孕的问题和解决办法

Problems 问题	Solutions 解决办法
Low sperm count 精子数量少	Injecting sperm cell into egg cell, then IVF (书上已给)
Oviducts 太窄或受损	doctors try to widen the oviducts; or IVF
Sperm ducts 堵塞	doctors try to widen the sperm ducts
内膜阻止 implantation	IVF
排出的成熟卵太少	IVF
Sperm 游不好	inject a sperm cell into an egg cell with a fine needle

1. doctors try to widen the oviducts; or IVF
2. doctors try to widen the sperm ducts
3. IVF
4. IVF

5. inject a sperm cell into an egg cell with a fine needle

Short solution:

These are the answers in the paragraphs on page 31. The numbers 1 to 5 are the five empty rows, from the top.

IVF：在培养皿里受精

把卡片排好顺序。

A 从女性的 ovaries 里取出 eggs。

B 把一个 embryo 放进 uterus，它可能在那里 implant。

C 给女性用 hormones，让多个 eggs 成熟。

D eggs 和 sperm 在培养皿里混合，其中一些受精。

E embryos 培养大约两天。

顺序：C → A → D → E → B

为什么 IVF 之后双胞胎更常见？

often more than one embryo is put into the uterus, and each one can implant

Teaching line, not a question slide.

Order: C, A, D, E, B.

Q often more than one embryo is put into the uterus, and each one

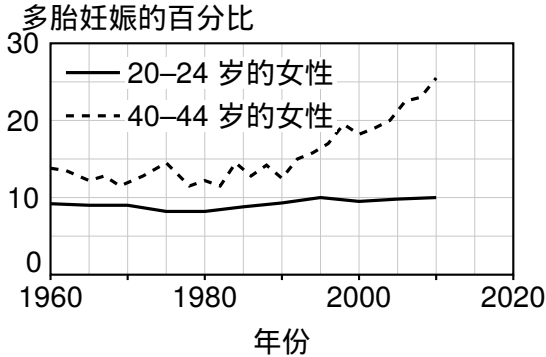
can implant

In vitro means in glass: fertilisation happens outside the body.

Say it:

vitro = VEE-troh

Workbook 练习册：第 3 题，describe



- (a) 描述 40–44 岁女性多胎妊娠数量的变化。
from 1960 to about 1990 it stayed between about 11 % and 15 %; after 1990 it rose steadily to about 25 % in 2010
- (c) 描述 20–24 岁女性多胎妊娠数量的变化。
it stayed almost the same, at about 9 % to 10 %, from 1960 to 2010

Q3 (a) from 1960 to about 1990 it stayed between about 11 % and 15 %; after 1990 it rose steadily to about 25 % in 2010
Q3 (c) it stayed almost the same, at about 9 % to 10 %, from 1960 to

2010
Short solution:
A multiple birth is two or more babies born from one pregnancy.

Workbook 练习册: suggest 和 predict

新闻标题：“生育治疗会增加多胎的机会” · “年龄大的女性更容易有怀孕困难” · “生育治疗越来越普遍”

Q3 (b) 提出一个假设 (hypothesis), 解释为什么会出现这种变化。
fertility treatment has become more common; older women are more likely to need it; fertility treatment increases the chance of a multiple birth

Q3 (d) 提出一个假设, 解释为什么 20–24 岁的规律和 40–44 岁不一样。
women aged 20 to 24 rarely have problems becoming pregnant, so few of them have fertility treatment

Q4 (a) 预测去年 40–44 岁女性多胎妊娠的百分比。
about 30 % or a little more: any value above 25 % that continues the rising line. (b) I extended the trend line from 1990 to 2010 forward to last year

Q3 (b) fertility treatment has become more common; older women are more likely to need it; fertility treatment increases the chance of a multiple birth

Q3 (d) women aged 20 to 24 rarely have problems becoming pregnant, so few of them have fertility treatment

Q4 (a) about 30 % or a little more: any value above 25 % that continues the rising line

Q4 (b) I extended the trend line from 1990 to 2010 forward to last year

Say it:

hypothesis = hy-POTH-uh-sis

Spot the error 找错 (4)

题目：描述胎儿怎样得到 oxygen，以及它在 uterus 里怎样受到保护。

Sam's answer

The mother's blood flows through the umbilical cord into the foetus. Oxygen moves by diffusion from a low concentration to a high concentration. The amniotic fluid gives the foetus its food and protects it from bumps.

Sam 的答案有 3 处错。找出来，改正，并说出为什么丢分。

1. the two bloods never mix: only the foetus's blood flows in the umbilical cord (L9)
2. diffusion is from a high concentration to a low concentration (L10)
3. the amniotic fluid does not feed the foetus: food comes from the mother's blood through the placenta (L13)

1. the two bloods never mix: only the foetus's blood flows in the umbilical cord (L9)
2. diffusion is from a high concentration to a low concentration (L10)
3. the amniotic fluid does not feed the foetus: food comes from the mother's blood through the placenta (L13)

Short solution:

Oxygen diffuses from the mother's blood into the foetus's blood in the placenta, from a high concentration to a low concentration. The foetus's blood carries it along the umbilical cord. The amniotic fluid protects the foetus from bumps.

Practice 练习

Q1 说出两种从胎儿的血进入母亲的血的物质。 [A]

carbon dioxide and urea

Q2 解释为什么吸烟的孕妇更可能生下偏小的婴儿。 [M]

carbon monoxide from the smoke is carried in the mother's blood, so less oxygen crosses the placenta; the foetus's cells respire less and release less energy, so it grows more slowly

Q3 现在做 IVF 通常只放一个 embryo 进 uterus，而不是三个。说出可能的原因，并解释这怎样改变双胞胎和三胞胎的数量。 [E]

a multiple birth is riskier for the mother and the babies, who are often premature and small; with one embryo only one can implant, so there are fewer twins and triplets

Q1 carbon dioxide and urea

Q2 carbon monoxide from the smoke is carried in the mother's blood, so less oxygen crosses the placenta; the foetus's cells

respire less and release less energy, so it grows more slowly

Q3 a multiple birth is riskier for the mother and the babies, who are often premature and small; with one embryo only one can implant, so there are fewer twins and triplets

In-class work · Part D (约 6 分钟)

- 1 给子宫里的 fetus 图标注。
- 2 解释一种废物怎样离开胎儿。
- 3 描述并解释一张关于 fetus 的数据表。

要写出：什么物质、往哪个方向、什么过程。

答案在 inclass_answers.pdf 的 Part D

Ways to lose marks 失分点

说出每一句的正确版本。

- 1 “A sperm cell has 46 chromosomes.”
23: every sex cell has one set; the zygote and body cells have 46 (L1)
- 2 “Fertilisation happens in the uterus.”
in the oviduct; the embryo implants in the uterus later (L2)
- 3 “Ovulation is when the lining is lost.”
ovulation is the release of an egg; losing the lining is menstruation (L5)
- 4 “The mother’s blood flows into the foetus.”
the bloods never mix; oxygen and food diffuse across the placenta (L9)
- 5 “A sperm has a tail.” (题目要求解释 adaptation)
add how it helps: so that it can swim to the egg (L3)
- 6 “It went up because of IVF.” (题目要求 describe 图表)
describe needs the trend and numbers; the reason is for suggest or explain (L12)

Teaching line, not a question slide.

1. 23: every sex cell has one set; the zygote and body cells have 46 (L1)
2. in the oviduct; the embryo implants in the uterus later (L2)
3. ovulation is the release of an egg; losing the lining is

menstruation (L5)

4. the bloods never mix; oxygen and food diffuse across the placenta (L9)
5. add how it helps: so that it can swim to the egg (L3)
6. describe needs the trend and numbers from the graph; the reason is for suggest or explain (L12)

- 1 Sex cells 有 23 条染色体；在 oviduct 里 fertilisation 之后是 46 条。
- 2 周期：period，内膜 builds up，第 14 天 ovulation，内膜保持 thick。
- 3 Placenta：diffusion，从高浓度到低浓度；两边的血永远不混合。
- 4 Explain = fact + because。Describe a graph = trend + numbers。

In-class work · Part E

写 12 分钟，不看笔记，不看 booklet。

然后一起批改：每道题给 A、M 或 E。

先看指令词：state、describe、explain、suggest。

答案在 inclass_answers.pdf 的 Part E

Homework: 两部分，每部分约 40 分钟

- 1 Part 1: 性细胞、男女生殖系统、月经周期。
- 2 Part 2: 受精、染色体、怀孕、不孕。

两部分分两天做。

答案在 homework_answers.pdf